Unit 3 · Outcome 1 · Maintaining Biodiversity

Threats to Biodiversity

KK10 VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 10 Why do species slip toward extinction? A species rarely vanishes for a single reason

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.

KEY KNOWLEDGE 10

Why do species slip toward extinction?

A species rarely vanishes for a single reason. It is squeezed by a web of pressures — some caused by people, some not — that interact and compound. KK10 is about naming those pressures precisely and explaining the mechanism by which each one reduces a population's size, range or genetic health.

Across this module our running case study is Kestrel Ridge Reserve — a fictional but realistic Victorian wetland modelled on Yellingbo, the last refuge of the critically endangered Helmeted Honeyeater. We will watch the same bird face every threat in the VCAA list, because real species are almost never threatened by just one thing.

◈ **ANALOGY**

Hold this analogy A threatened species is like a patient with several illnesses at once. You can't treat just one and call the job done — and the illnesses make each other worse. A small population (one illness) makes inbreeding more likely (a second), which lowers disease resistance (a third). Examiners reward students who explain these interactions, not just lists.

◆ **NOTE**

Define it cleanly Threat to biodiversity: any process that reduces the genetic, species or ecosystem diversity of a region — usually by shrinking populations, reducing a species' geographic range, or lowering its genetic variation. Human (anthropogenic) threat: caused or accelerated by human activity (e.g. land clearing, pollution, introduced species). Non-human threat: arises without direct human cause (e.g. a naturally occurring disease, a drought). Many threats are both — climate change is natural in origin but is being driven faster by humans.

ORIENTATION

The threat map

Eight pressures sit behind the VCAA dot point. Learn them as a set, tag each as predominantly human or non-human, and — crucially — learn the one-line mechanism for each. Hover a card; the mechanism is what earns marks.

Clearing, draining or fragmenting habitat removes the resources a species needs and splits one large population into small isolated ones.

Harvesting, hunting or collecting faster than a population can replace itself drives numbers down and shrinks range.

In a small population, related individuals breed; harmful recessive alleles pair up and genetic diversity is lost.

Losing pollinators, seed dispersers, hosts or symbionts breaks the relationships a species needs to reproduce or persist.

Persistent pollutants build up in organisms (bioaccumulation) and concentrate up the food chain (biomagnification), poisoning top predators.

Shifting temperature, rainfall and fire regimes move habitats faster than species can track, and intensify extreme events.

Pathogens can sweep through a population — especially one already small, stressed or genetically uniform.

Non-native predators and competitors take shelter, food and water from native species that did not evolve alongside them.

✓ **TIP**

Exam framing When a question says " describe two threats " it wants a named threat + its mechanism + an effect on the species. "Habitat loss" alone is barely worth a mark. " Habitat loss removes the dense swamp vegetation the honeyeater nests in, so fewer pairs can breed and the population declines " is a full answer.

THREAT 1 & 2

Habitat loss & the small-population trap

⚠ **EXAM TRAPS**

Habitat loss and over-exploitation are the two engines that create small, isolated populations — and small isolated populations are where most other threats do their damage. Use the explorer below to watch Kestrel Ridge fragment and the honeyeater population thin out. Three related ideas often get muddled. Keep them distinct:

Term	What happens	Why it threatens the species
Habitat loss	Total area of suitable habitat falls (e.g. swamp drained for farmland).	Fewer resources → the area can support fewer individuals (lower carrying capacity).
Fragmentation	Remaining habitat is broken into disconnected patches.	Patches are too small to be self-sustaining; movement between them is risky.
Isolation	Sub-populations can no longer interbreed across patches.	No gene flow → each patch drifts genetically and is vulnerable to local extinction.

◆ **NOTE**

▲ Over-exploitation counts too For the honeyeater, over-exploitation is minor — but for many species (timber trees, fish, abalone, some orchids) harvesting faster than the replacement rate is the primary threat. The mechanism is the same: numbers fall below the level where the population can recover, range contracts, and the survivors become a small, vulnerable group.

◈ **ANALOGY**

Analogy · islands Fragmentation turns a continent into a scattering of islands. A small island can't hold a big stable population — a single bad year (fire, flood, disease) can wipe it out, and there's no easy way for animals from a neighbouring island to recolonise. That's why isolation is so dangerous: it removes the rescue effect.

THREAT 3

Inbreeding & the extinction vortex

Once a population is small, a genetic trap snaps shut. Inbreeding raises the chance that offspring inherit two copies of a harmful recessive allele, while genetic diversity — the raw material for adapting to change — drains away. Watch it happen in the bottleneck simulator.

◆ **NOTE**

Two distinct problems Inbreeding depression: mating between close relatives increases homozygosity, so harmful recessive alleles are expressed — reducing fertility, survival and overall fitness. Loss of genetic diversity: a small population carries fewer alleles to begin with, and genetic drift removes more each generation. With less variation, the species is less able to adapt to new diseases, predators or climate.

These effects feed back on one another, dragging a population downward in a self-reinforcing spiral:

⚠ **WATCH OUT**

Common trap Don't write "inbreeding causes mutations". It doesn't create new harmful alleles — it raises the chance that existing recessive alleles pair up and are expressed. Precision here separates a 4/5 from a 5/5.

THREAT 4

Lost partners — pollinators, dispersers, hosts & symbionts

No species lives alone. Many depend on a partner to reproduce or persist. The VCAA wording is specific: loss of pollinators, dispersal agents, host species or symbionts that affect reproduction and persistence. Lose the partner and you threaten the species — even if nothing touches it directly.

Insects, birds or bats that move pollen so plants can set seed. No pollinator → no seed → no next generation of plants (and the animals that feed on them).

Animals that carry seeds away from the parent (in gut or fur). Without them, seedlings crowd the parent and the plant can't colonise new ground.

A species that another requires to complete its life cycle — e.g. a butterfly whose caterpillars eat only one plant. Lose the host, lose the dependent.

Partners in a mutually beneficial relationship — e.g. mycorrhizal fungi that feed tree roots, or gut microbes. Break the partnership and both decline.

◈ **ANALOGY**

Analogy · the keystone in an arch Some partners are keystone relationships: remove one and a whole structure of dependent species collapses. A fig tree that feeds dozens of animals, or a fungus that lets every tree in a forest absorb nutrients — losing it triggers a cascade far larger than the single species lost.

✓ **TIP**

Link to Kestrel Ridge The honeyeater feeds on nectar and lerp (sap-sucking insect secretions). It also pollinates the very plants it feeds from. Lose the bird and the plants lose a pollinator; lose the plants and the bird loses its food. This mutual dependence is why protecting one species so often means protecting a relationship.

THREAT 5 · HIGH-FREQUENCY EXAM TOPIC

Bioaccumulation & biomagnification

This is the most mixed-up pair of terms in the whole course — and examiners love testing whether you can tell them apart. Build the tower below and watch a persistent pollutant climb a food chain.

◆ **NOTE**

Bioaccumulation Build-up of a persistent pollutant within a single organism over its lifetime, because it is absorbed faster than it can be broken down or excreted. (Think: one fish, getting more contaminated as it ages.)

◆ **NOTE**

Biomagnification Increase in pollutant concentration from one trophic level to the next, up a food chain. A predator eats many contaminated prey, so it carries a higher concentration than any single prey did. (Think: across the chain, not within one body.)

Feature	Bioaccumulation	Biomagnification
Where	Within one organism	Between trophic levels
Over what	The organism's lifetime	Along a food chain
Requires	A persistent, fat-soluble toxin	Bioaccumulation + a food chain
Worst hit	Long-lived individuals	Top predators (e.g. the honeyeater, raptors)

◆ **NOTE**

▲ Persistence is the key word Only persistent pollutants biomagnify — substances that resist breakdown and are stored in fat (e.g. DDT, mercury, PCBs). A toxin that the body quickly excretes does not accumulate or magnify. If a question gives you a chemical, ask: is it persistent and fat-soluble?

⚠ **WATCH OUT**

The classic mistake "Biomagnification is when the toxin builds up in an animal over time." ✗ — that's bioaccumulation. Biomagnification is the jump between trophic levels. Whenever you use one term, define the other to show you know the difference.

THREATS 6 · 7 · 8

Climate change, disease & introduced species

The final three threats often act as multipliers — they hit hardest when a population is already small, fragmented or genetically uniform from the threats above.

Shifting temperature and rainfall move the climate envelope a species is adapted to. Species must shift range, adapt, or decline — and a fragmented landscape blocks the shift. For Kestrel Ridge, hotter, drier summers raise bushfire risk (the 2009 fires came within kilometres of Yellingbo) and dry out the swamp the honeyeater needs.

A pathogen can crash a population fast — and a genetically uniform population (from inbreeding) has little variation in disease resistance, so few individuals survive. Disease is listed as both human and non-human: emergence is often natural, but spread is frequently aided by human movement and introduced carriers.

The VCAA wording is exact: introduced species that compete for shelter, food and water — plus predation and disease. Native species did not evolve alongside the invader and lack defences. At Yellingbo, Bell Miners and Noisy Miners aggressively exclude the Helmeted Honeyeater, while foxes and cats prey on nests.

Invader impact	Mechanism	Example
Competition	Takes shelter, food or water natives rely on	Rabbits stripping ground cover; miners excluding small birds
Predation	Eats natives that have no evolved defence	Foxes & cats on ground-nesting birds
Disease / parasites	Introduces pathogens natives can't resist	Chytrid fungus in frogs
Habitat change	Alters the habitat structure itself	Carp muddying waterways; weeds choking swamp

✓ **TIP**

Synthesis The strongest exam answers show interaction: "Habitat loss creates a small, isolated honeyeater population (threat 1). That population inbreeds, lowering disease resistance (threats 3 + 7), while introduced Bell Miners exclude it from remaining habitat (threat 8) and climate-driven fire threatens its last swamp (threat 6)." That single sentence touches five threats and their interactions.

BUILD SKILL · CAUSE → MECHANISM → EFFECT

Build a causal chain

Top-band answers always trace a chain: a named threat, the mechanism it works through, and the measurable effect on the species. Pick a threat below to see a worked chain, then try writing your own in the Writing Lab.

WRITING LAB

Command words, P·E·L & exam traps

⚠ **EXAM TRAPS**

Knowing the content isn't enough — you must answer the verb. Tap a command word to see what the examiner actually wants, then drill the P·E·L structure on a real prompt. A reliable structure for short-answer extended responses: Point (state it), Explain (the mechanism), Link (back to the species/question). Draft each line, then reveal a model. Point: Fragmentation breaks the honeyeater's swamp habitat into small, isolated patches. Explain: Each patch supports only a few breeding pairs, and because birds rarely cross the cleared gaps between patches there is little gene flow — so the sub-populations inbreed, lose genetic diversity, and any single patch can be wiped out by one fire or disease outbreak. Link: With low diversity and no rescue from neighbouring patches, the species' ability to adapt and recover falls, pushing it toward extinction over the long term.

⚠ **WATCH OUT**

Trap 1 · listing without mechanism Naming threats ("habitat loss, climate change, foxes") earns almost nothing on "explain/analyse" questions. Always add how each one reduces population, range or diversity.

⚠ **WATCH OUT**

Trap 2 · bioaccumulation ≠ biomagnification Within one body = bioaccumulation. Up the food chain = biomagnification. Define both whenever you use either.

⚠ **WATCH OUT**

Trap 3 · "threats" vs "threatened" A threat is a process (e.g. land clearing). Threatened is a conservation status (KK08). Don't confuse the two — a question about threats wants processes, not categories.

⚠ **WATCH OUT**

Trap 4 · inbreeding "creates" mutations Inbreeding exposes existing recessive alleles; it doesn't generate new ones. Say "increases homozygosity / expression of harmful recessives".

PRACTICE EXAM · SELF-MARK

Five VCAA-style questions

★ **PRACTICE & SELF-MARK**

Attempt each question before revealing the responses. Compare the weak and strong answers, read the marking guide, then mark yourself honestly using the buttons. Your total tracks in the dock and saves automatically.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.